DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW



Evaluation Scheme & Syllabus

For

B.Tech. 2nd Year

Aeronautical Engineering

(Effective from the Session: 2023-24)

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

SEMESTER –III

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BOE3** / BAS303	Science Based Open Elective/BSC (Maths-III/Math IV/ Math V)	T	ES/BS	3	1	0	20	10	30	70	100	4
2	BVE301 / BAS301	Universal Human Value and Professional Ethics/ Technical Communication	T	VA/HS	2	1	0	20	10	30	70	100	3
3	BAE301	Fundamental of Thermodynamics	T	PC	3	1	0	20	10	30	70	100	4
4	BME302	Fluid Mechanics & Fluid Machines	T	PC	3	1	0	20	10	30	70	100	4
5	BME303	Materials Engineering	T	PC	2	1	0	20	10	30	70	100	3
6	BME351	Fluid Mechanics Lab	P	PC	0	0	2		50	50	50	100	1
7	BME352	Material Testing Lab	P	PC	0	0	2		50	50	50	100	1
8	BME353	Computer Aided Machine Drawing-I Lab	P	PC	0	0	2		50	50	50	100	1
10	BCC301 / BCC302	Cyber Security/Python programming	T	VA	2	0	0	20	10	30	70	100	2
11	BCC351	Internship Assessment /Mini Project*	P							100		100	2
		Total			**15**	**5**	**6**						**25**

- **Mathematics –III** for CE / ENV and allied branches
- **Mathematics-IV** for Computer/Electronics/Electrical & allied Branches, Mechanical & Allied Branches Textile/Chemical & allied Branches
- **Mathematics-V** for Bio Technology / Agriculture Engineering

SEMESTER –IV

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BAS403 / BOE4**	BSC (Maths-III/Math IV/ Math V)/Science Based Open Elective	T	BS/ES	3	1	0	20	10	30	70	100	4
2	BAS401 / BVE401	Technical Communication / Universal Human Value and Professional Ethics	T	HS/VA	2	1	0	20	10	30	70	100	3
3	BAE401	Introduction to Aeronautical Engineering	T	PC	3	1	0	20	10	30	70	100	4
4	BAE402	Mechanics of Solids	T	PC	3	1	0	20	10	30	70	100	4
5	BME403	Manufacturing Processes	T	PC	2	1	0	20	10	30	70	100	3
6	BME451	Applied Thermodynamics Lab	P	PC	0	0	2		50	50	50	100	1
7	BME452	Manufacturing Processes Lab	P	PC	0	0	2		50	50	50	100	1
8	BME453	Computer Aided Machine Drawing-II Lab	P	PC	0	0	2		50	50	50	100	1
9	BCC402 / BCC401	Python Programming/Cyber Security	P	VA	2	0	0	20	10	30	70	100	2
10	BVE451 / BVE452	Sports and Yoga - II / NSS-II	P	VA	0	0	3			100		100	0
		Total			**15**	**5**	**9**						**23**
		Minor Degree/ Honors Degree MT-1/HT-1											

***The Mini Project or internship (4 weeks) will be done during summer break after 4th Semester and will be assessed during V semester.

BAE301- FUNDAMENTAL OF THERMODYNAMICS

Course Outcomes: After completion of this course, the students will be able to		Bloom's Taxonomy
CO1	Understand the systems and control volume in situations involving heat and work interactions and changes in thermodynamic properties of substances.	**K2**
CO2	Understand the thermodynamic laws and evaluate the performance of energy conversion devices.	**K3**
CO3	Apply the laws of thermodynamic, entropy and available and unavailable energy in real life application.	**K3**
CO4	Understand the differences between high-grade and low-grade energies and changes in thermodynamic properties of substances.	**K2**
CO5	Understand the construction and working of boilers and condensers and evaluate the boiler parameters.	**K3**

UNIT-I (08 Hours)

Fundamental Concepts and Definitions: Introduction and basic concepts: system, control volume, surrounding, boundaries, universe, types of systems, macroscopic and microscopic viewpoints, concept of continuum, thermodynamic equilibrium, state, property, process, exact & inexact differentials, cycle reversibility quasi-static process, irreversible process, causes of irreversibility, energy and its forms, work and heat (sign convention), Helmholtz & Gibb's function.

UNIT-II (08 Hours)

Zeroth law of thermodynamics: Concept of temperature scales and its measurements.

First law of thermodynamics: Thermodynamic definition of work, displacement work and flow work, displacement work for various non-flow processes, Joule's experiment, first law analysis for closed system (non-flow processes), internal energy and enthalpy, limitations of first law of thermodynamics, PMM-1 and its applications, steady flow systems and their analysis, steady flow energy equation, boilers, condensers, turbine, throttling process, pumps.

UNIT–III (10 Hours)

Second law of thermodynamics: Thermal reservoirs, energy conversion, heat engines, efficiency, reversed heat engine, heat pump, refrigerator, coefficient of performance, Kelvin-Plank and Clausius statements and their equivalence, Carnot's theorem, Carnot cycle and Carnot engine, PMM-2.

Entropy: Clausius inequality, concept of entropy, entropy change of pure substance in different thermodynamic processes, T-ds equation, principle of entropy increase, T-s diagram, statement of the third law of thermodynamics, available and unavailable energy, availability and irreversibility, second law efficiency.

UNIT–IV (08 Hours)

Properties of steam: Property of pure substance (steam), triple point, critical point, saturation states, sub-cooled liquid state & superheated vapor state, phase transformation process of water, p-T & p-v diagrams, T-s and h-s diagrams, use of property diagram, dryness factor and its measurement, processes involving steam in closed and open systems.

Rankine cycle: Principle of Rankine cycle, working, application and limitations of Rankine cycle.

UNIT–V (08 Hours)

Boilers: Classifications and working of boilers, air pre-heater, feed water heater, super heater. boiler efficiency, equivalent evaporation, boiler trial and heat balance.

Draught and its calculations: Working principle of draught, types of draught, evaluation of draught height, diameter and other parameters.

Condenser: Working principle, classification, performance and other parameters of condenser.

Books and References:

1. Engineering Thermodynamics by P.K.Nag, TMH
2. Thermodynamics-An Engineering Approach by Cengel & Boles, TMH.
3. Fundamentals of Engineering Thermodynamics by Rathakrishnan, PHI.
4. Engineering Thermodynamics by Dhar, Elsevier.
5. Engineering Thermodynamics by Onkar Singh, New Age International.
6. Engineering Thermodynamics by R.K. Rajput, S. Chand Publications.
7. Engineering Thermodynamics by C.P. Arora.
8. Engineering Thermodynamics by Rogers and Mayhew, Pearson.

BME302- FLUID MECHANICS AND FLUID MACHINES

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Understand the application of mass and momentum conservation laws for fluid flows.	K2
CO 2	Understand the importance of dimensional analysis.	K2
CO 3	Evaluate the velocity and pressure variations in various types of simple flows.	K3
CO 4	Mathematically analyze the flow in water pumps and turbines.	K3
CO 5	Understand about the functioning of centrifugal and reciprocating pumps.	K2
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Definition of fluid, Newton's law of viscosity, Units and dimensions-Properties of fluids, mass density, specific volume, specific gravity, viscosity, compressibility and surface tension, Incompressible flow, Bernoulli's equation and its applications - Pitot tube, orifice meter, and venturi meter.	08
II	Continuum & free molecular flows. Steady and unsteady, uniform and non-uniform, laminar and turbulent flows, rotational and irrotational flows, compressible and incompressible flows, subsonic, sonic and supersonic flows, sub-critical, critical and supercritical flows, one, two- and three dimensional flows, streamlines, continuity equation for 3D and 1D flows.	08
III	Equation of motion for laminar flow through pipes resistance to flow, minor losses, pipe in series and parallel, power transmission through a pipe, siphon, water hammer. Boundary layer thickness, boundary layer over a flat plate, laminar boundary layer, application of momentum equation, turbulent boundary layer, laminar sublayer, separation and its control, Drag and lift.	08
IV	Introduction to hydrodynamic thrust of jet on a fixed and moving surface, Classification of turbines, Impulse turbines, Constructional details, Velocity triangles, Power and efficiency calculations, Governing of Pelton wheel. Francis and Kaplan turbines, Constructional details, Velocity triangles, Power and efficiency Selection of water turbines.	08
V	Classifications of centrifugal pumps, Vector diagram, Work done by impellor, Efficiencies of centrifugal pumps, Specific speed, Cavitation & separation, reciprocating pump theory, Slip, Indicator diagram, Effect of acceleration, air vessels, Comparison of centrifugal and reciprocating pumps.	08

Books and References:

1. Introduction to fluid mechanics and Fluid machines by S.K Som, Gautam Biswas, S Chakraborty.
2. Fluid mechanics and machines by R.K Bansal.
3. F. M. White, Fluid Mechanics, 6th Ed., Tata McGraw-Hill, 2008.
4. Fluid Mechanics and Its Applications by V.K. Gupta et.al. 5. Fluid Mechanics by Yunus Cengel.
5. Batchelor, G. K. (1999). Introduction to fluid dynamics. New Delhi, India: Cambridge University Press.
6. Acheson, D. J. (1990). Elementary fluid dynamics. New York, USA: Oxford University Press.
7. R.W. Fox, A.T. McDonald and P.J. Pritchard, Introduction to Fluid Mechanics, 6th Ed., John Wiley, 2004.
8. Fluid Mechanics and Hydraulic Machines, 5/ed by Mahesh Kumar, Pearson Education
9. Fluid Mechanics, 2e in SI Units, by R.C Hibbeler, Pearson Education

BME303- MATERIALS ENGINEERING

CO-1: Students will be able to identify the crystal structure and measure the mechanical properties of materials. **K3**
CO-2: Students will be able to test the various failures of materials. **K3**
CO-3: Students will be able to identify the mechanical properties based on composition of micro-constituents depicted in the phase-diagram. **K3**
CO-4: Students will understand the concept of improving the mechanical properties through heat treatment. **K2**
CO-5: Students will learn the structure and properties of alloys and composites. **K2**

UNIT-I (08 Lecture)

Crystal Structure: Unit cells, Metallic crystal structures, Ceramics. Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, critically resolved shear stress. Mechanical Properties and its measurement: Tensile, compression and torsion tests; Young's modulus, relations between true and engineering stress-strain curves, generalized Hooke's law, yielding and yield strength, ductility, resilience, toughness and elastic recovery; Hardness: Rockwell, Brinell and Vickers and their relation to strength.

UNIT-II (08 Lecture)

Alloys, substitutional and interstitial solid solutions- Hume-Rothery Rule, Phase diagrams: Interpretation of binary phase diagrams and micro-structure development; Lever Rule, eutectic, eutectoid, peritectic, peritectoid and monotectic reactions. Iron and Iron-carbide phase diagram, microstructural aspects of ledeburite, austenite, ferrite and cementite, cast iron, High Speed Steel

UNIT-III (08 Lecture)

Alloying of steel, properties of stainless steel and tool steels, maraging steels- cast irons; grey, white, malleable and spheroidal cast irons-copper and copper alloys; brass, bronze and cupro-nickel; Aluminium and Al-Cu – Mg alloys- Nickel based superalloys and Titanium alloys, Carbon Fibre and alloys, Structure and properties of Graphene Material, Composite Materials, Reinforcement

UNIT-IV (08 Lecture)

Heat treatment of Steel: Annealing, tempering, normalising and spheroidising, isothermal transformation diagrams for Fe-C alloys and microstructure development. Continuous cooling curves and interpretation of final microstructures and properties austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

UNIT-V (08 Lecture)

Creep and failures, Introduction to non-destructive testing (NDT): Visual Inspection, Liquid Penetrate Testing, Magnetic Particle Inspection, Ultra-sonic Testing, Acoustic Testing, Radiography, X-ray Differaction, Eddy Current Testing etc., Material behaviour / properties at a very low (Cryogenics) and high temperature.

Books and References:

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. Raghavan, "Material Science and Engineering', Prentice Hall of India Private Limited, 1999.
4. Mechanics of materials by James M. Gere.
5. Introduction to engineering materials by B.K. Agarwal.
6. Physical metallurgy and advanced materials by R.E. Smallman.
7. Engineering mechanics of composite materials by Isaac M. Daniel.
8. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
9. Introduction to Materials Science For Engineers by Shackelford, 9th edn, 2023, Pearson Education
10. Engineering Materials: Properties and Selection, 9edn by Budinski, Pearson Education

BME351- FLUID MECHANICS LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand the principles and performance characteristics of flow and thermal devices.	**K2**
CO 2	Know about the measurement of the fluid properties	**K1**
CO 3	Understand and analyze various properties of fluids	**K3**
CO 4	Evaluate the performance characteristics of fluid/thermal machinery	**K3**
CO 5	Evaluate the velocity and pressure variations in various types of simple flows.	**K3**
	DETAILED SYLLABUS	**0-0-2**

List of Experiments:(At least 8 of the following)

1. To determine the coefficient of impact for vanes.
2. To determine coefficient of discharge of an orifice meter.
3. To determine the coefficient of discharge of Notch (V and Rectangular types).
4. To determine the friction factor for the pipes.
5. To determine the coefficient of discharge of venturi meter.
6. To determine the coefficient of discharge, contraction & velocity of an orifice.
7. To verify the Bernoulli's Theorem.
8. To find critical Reynolds number for a pipe flow.
9. To determine the meta-centric height of a floating body.
10. To determine the minor losses due to sudden enlargement, sudden contraction and bends.
11. To show the velocity and pressure variation with radius in a forced vortex flow.

BME352- MATERIALS TESTING LAB

CO-1: Students will be able to perform different destructive and non-destructive testing methods to measure various mechanical properties. **(K2)**
CO-2: Students will be able to analyse the effect of different heat-treatment processes on the Hardness. **(K4)**
CO-3: Students will be able to simulate the material using simulating software / measure the mechanical properties of 3-D printed components. **(K3)**

List of Experiments: (At least 8 of the following)

1. Strength test of a given mild steel specimen on UTM with full details and stress versus strain plot on the machine.
2. Other tests such as shear, bend tests on UTM.
3. Impact test on impact testing machine like Charpy, Izod or both.
4. Hardness test of given specimen using Rockwell and Vickers/Brinell testing machines (this test should be conducted on MS Specimen – prior and after the various heat-treatment processes like Annealing / Normalizing / Hardening etc.; with this, 2 experiments can be designed: one for Annealing or Normalizing and another for Hardening / Tempering).
5. Fatigue test on fatigue testing machine.
6. Creep test on creep testing machine.
7. Study of NDT (non-destructive testing) methods like magnetic flaw detector, ultrasonic flaw detector, eddy current testing machine, dye penetrant tests.
8. Modelling of Graphene / any other materials structure using LAMMPs (LAMMPS Molecular Dynamics Simulator) or any other simulation software and studying the mechanical properties.
9. Measurement of mechanical properties (tensile test, compression test, hardness test / flexural test) of specimen made by 3-D printing

BME353- COMPUTER AIDED MACHINE DRAWING-I LAB

	Course Outcome (CO)	Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand and apply 2D software to develop a part model	**K3**
CO 2	Understand about temporary and permanent fasteners	**K2**
CO 3	Understand the need for free hand sketching, Free hand sketching of foundation bolts etc.	**K2**
CO 4	Create assembly drawing of simple machine elements like rigid or flexible coupling	**K3**
CO 5	Create 2D drawings and assemblies of various machine components	**K3**
	DETAILED SYLLABUS	**0-0-2**

UNIT-I

Computer aided drafting (1 drawing) Introduction to computer aided drafting; advantages and applications of CAD, concepts of computer aided 2D drafting using any drafting software like AutoCAD, Pro-E, Solid Edge, Draft Sight etc., basic draw and modify commands, making 2D drawings of simple machine parts. Course Outcomes: Upon completion of this course, the students can use computer and CAD software for modelling mechanical components.

UNIT-II

Fasteners (2 drawing sheets) Temporary and permanent fasteners, thread nomenclature and forms, thread series, designation, representation of threads, bolted joints, locking arrangement of nuts, screws, washers, foundation bolts etc., keys, types of keys, cotter and knuckle joints.

UNIT-III

Riveted joints (1 drawing sheet) Introduction, rivets and riveting, types of rivets, types of riveted joints, drawing of boiler joints etc. Free hand sketching (1 drawing sheet) Introduction, need for free hand sketching, Free hand sketching of foundation bolts, studs, pulleys, couplings etc.

UNIT-IV

Assembly drawing (2 drawing sheets) Introduction to assembly drawing, drawing assembly drawing of simple machine elements like rigid or flexible coupling, muff coupling, Plummer block, footstep bearing, bracket etc.

Books and References:

1. Fundamentals of Machine Drawing by Sadhu Singh & Shah, PHI.
2. Engineering Drawing by Bhat, & Panchal, Charotar Publishing House.
3. Machine Drawing with AutoCAD by Pohit and Ghosh, Pearson.
4. Machine Drawing-KL Narayana, P Kannaiah, KV Reddy, New Age.
5. Machine Drawing, N. Siddeshswar, P Kannaiah, VVS Shastry, Tata McGraw Hill.
6. Engineering Drawing, Pathak, Wiley.
7. Textbook of Machine Drawing, K C John, PHI.
8. AutoCAD 2014 for Engineers & Designers, Bhatt, WILEY
9. Engineering Drawing, 2nd edn by Shah & Rana
10. Fundamentals of Engineering Drawing, 11th edn by Warren J. Luzadder, Pearson Education

SEMESTER-IV

BAE401- INTRODUCTION TO AERONAUTICAL ENGINEERING

Course Outcomes: After completion of this course, the students will be able to		**Bloom's Taxonomy**
CO1	Understand the evolution of conventional aircraft and generation of lift.	**K2**
CO2	Understand the various forces and moments acting on an aircraft.	**K2**
CO3	Explain the different parts and working of aircraft engines.	**K2**
CO4	Understand the different materials used in the structural components of aircraft.	**K2**
CO5	Explain the working of different systems installed in aircraft and understand the aviation legislation.	**K2**

UNIT-I (08 Hours)

Introduction and History: History and evolution of aircraft, conventional airplane, progress in airplane design and application, current status, other kinds of heavier than air vehicle, helicopter, VTOL machines, space vehicles, reusable space vehicles and space shuttle, different parts of airplane, airfoil nomenclature and types, effect of viscosity and concept of boundary layer, high lift devices.

UNIT-II (08 Hours)

Basics of aerodynamics: Aerodynamic characteristics, types of lifting surface, types of wing plane- forms, lift and drag forces, lift to drag ratio as efficiency of a lifting surface, aspect ratio, airplane axis system, forces and moments, equilibrium of forces developed on wing and horizontal tail, aircraft stability, location of centre of gravity and centre of pressure their importance, elementary performance of airplane, use of elevator, rudder and ailerons.

UNIT-III (08 Hours)

Basics of propulsion: Requirement of power to fly, balance of forces, various means of producing power for forward flight, engine airframe compatibility, propellers nomenclature, types of propellers and theirs uses, different types of piston and jet engines, locations of such engines, rocket engines and various types of propellants.

UNIT-IV (09 Hours)

Basics of structure: Configuration of fuselage and wings, evolution of structure, progress in materials: wooden to all metal airplanes, strength to weight ratio of aircraft materials, importance of weight load factors of safety in aeronautics and aerospace applications, details of the structural layout of wing, fuselage tail planes, cockpit and cabin configuration, different types of materials for airplane and engine application, materials for space vehicles, advance composite materials.

UNIT-V (10 Hours)

Aircraft systems: Elementary studies on hydraulic, pneumatic, pressurizing air-conditioning and oxygen systems, landing gear and control surface actuating system, aircraft electrical systems, elementary studies of generation and on-board distribution of electricity.

Aviation legislation: Guidelines for airplane design, type certification and airworthiness certificate, role of DGCA in air safety and regulatory authority, accident investigation and rules of maintenance and services.

Books and References:

1. Fundamentals of Flight Richard S. Shevel, Prentice Hall.
2. Aircraft Basic Science: Ralph D. Bent & James I, Mackinley.
3. Jet Aircraft Power System: Jack V. Casamassa & Ralph D. Bent.
4. Aircraft Maintenance and Repair Kroes et al, GLENCOE, 1993.
5. Aircraft Aerodynamics: Clark B. Millikan, 1942
6. Introduction to Avionics: P.S. Dhunta, 1997.

BAE402- MECHANICS OF SOLIDS

Course Outcomes: After completion of this course, the students will be able to		**Bloom's Taxonomy**
CO1	Understand the significance of centroid and moment of inertia in structural components.	**K2**
CO2	Understand the nature of stresses developed in simple geometries like bars, beams, shafts, columns, cylinders and spheres for various types of simple loads.	**K2**
CO3	Determine the safe loads for components subjected to buckling.	**K3**
CO4	Evaluate the strain and deformation resulting from various simple and compound loading conditions.	**K3**
CO5	Determine the stresses acting on rotating shafts and curved beams.	**K3**

UNIT-I (08 Hours)
Force Systems: Concept of force, moment and couple; types of force systems; free body diagram, resultant of force systems: parallelogram law, polygon law, Lami's theorem, Varignon's theorem; principle of transmissibility of forces.
Properties of cross-section: centroid of plane figures; concept of moment of inertia, polar moment of inertia, product of inertia; parallel and perpendicular axis theorems; radius of gyration.

UNIT-II (08 Hours)
Simple stress system: types of stress: normal and shear; concept of strain, Hooke's law, Poisson's ratio; gradual, sudden and impact loads; strain energy; thermal stresses.
Compound stress system: Stresses on inclined planes in 2-D and 3-D; magnitudes and planes of principal stresses and maximum shear stress; Mohr's circle, principal strains; theories of failure.

UNIT-III (09 Hours)
Stresses in beams: Types of beams; determination of support reactions, dhear force diagram and bending moment diagram, pure bending, normal and transverse stresses in beams; combined bending and direct stresses; composite beams.
Deflection of beams: Double integration method, Macaulay's method and moment area method.

UNIT-IV (08 Hours)
Stresses in shafts: Torsional shear stresses in circular cross-section shafts; combined bending & torsion; shafts in series and parallel; stresses in rotating shaft.
Buckling of columns: Slenderness ratio; Euler's theory, effect of end conditions on column buckling; Rankine's theory.

UNIT-V (08 Hours)
Pressure Vessels: Types of stresses in thin-walled and thick-walled pressure vessels; strains in cylindrical pressure vessels with flat ends and hemi-spherical ends; spherical pressure vessels, built-in pressure vessels.
Bending of curved beams: Position of neutral axis for rectangular and circular sections, stresses in crane hooks, stresses in circular rings.

Books and References:

1. Engineering Mechanics, K.L.Kumar, Tata McGraw Hill
2. Strength of Materials, S.S.Rattan, Tata McGraw Hill
3. Mechanics of Materials, Beer & Johnston, Tata McGraw Hill
4. Mechanics of Solids, Bhavikatti, New Age International
5. Strength of Materials, R.K. Rajput, S.Chand & Company

BME403- MANUFACTURING PROCESSES

CO-1 : Students will learn the various conventional manufacturing processes / casting and forming processes. **K2**
CO-2 : Students will understand the concepts of metal cutting and CNC machining. **K2**
CO-3 : Students will comprehend the knowledge of grinding and super finishing processes. **K2**
CO-4 : Students will understand the concepts of metal joining processes. **K2**
CO-5 : Students will learn the concepts of unconventional machining processes. **K2**

UNIT-I (08 Lecture)
Conventional Manufacturing processes:
Casting and moulding: Metal casting processes and equipment, Types of Patterns and Pattern Allowance, Heat transfer and solidification, shrinkage, riser design, casting defects and residual stresses. Introduction to bulk and sheet metal forming, plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk forming (forging, rolling, extrusion, drawing) and sheet forming (shearing, deep drawing, bending), principles of powder metallurgy.

UNIT-II (08 Lecture)
Metal cutting:
Single and multi-point cutting; Orthogonal cutting, various force components: Chip formation, Tool wear and tool life, Surface finish and integrity, Machinability, cutting tool materials, cutting fluids, Coating; Turning, Drilling, Milling and finishing processes,
Introduction to CNC machining, Additive manufacturing, Rapid prototyping and rapid tooling

UNIT-III (08 Lecture)

Grinding & Super finishing:
Grinding: Grinding wheels, abrasive & bonds, cutting action. Grinding wheel specification. Grinding wheel wear - attritions wear, fracture wear. Dressing and Truing. Max chip thickness and Guest criteria. Surface and cylindrical grinding. Centreless grinding.
Super finishing: Honing, lapping and polishing.

UNIT-IV (08 Lecture)
Metal Joining (Welding):
Joining/fastening processes: Physics of welding, brazing and soldering; design considerations in welding, Solid and liquid state joining processes, Survey of welding and allied processes., Gas welding and cutting, process and equipment. Arc welding: Power sources and consumables. TIG & MIG processes and their parameters. Resistance welding - spot, seam projection etc. Other welding processes such as atomic hydrogen, submerged arc, electroslag, friction welding. Weld decay in HAZ.

UNIT-V (08 Lecture)
Unconventional Machining Processes:
Abrasive Jet Machining, Water Jet Machining, Abrasive Water Jet Machining, Ultrasonic Machining, principles and process parameters. Electrical Discharge Machining, principle and processes parameters, MRR, surface finish, tool wear, dielectric, power and control circuits, wire EDM; Electrochemical machining (ECM), Laser Beam Machining (LBM), Plasma Arc Machining (PAM) and Electron Beam Machining.

Books and References:

1. Kalpakjian and Schmid, Manufacturing processes for engineering materials (5th Edition)- Pearson India, 2014.
2. Mikell P. Groover, Fundamentals of Modern Manufacturing: Materials, Processes, and Systems.
3. Manufacturing Technology by P.N. Rao., MCGRAW HILL INDIA.
4. Materials and Manufacturing by Paul Degarmo.
5. Manufacturing Processes by Kaushish, PHI.
6. Principles of Foundry Technology, Jain, MCGRAW HILL INDIA
7. Production Technology by RK Jain.
8. Degarmo, Black &Kohser, Materials and Processes in Manufacturing.

BME451- APPLIED THERMODYNAMICS LAB

Objectives:

CO1: To understand the principles of various boilers: **K2**

CO2: To understand the basic principles IC engines and determination of various performance parameters of IC Engines: **K3**

CO3: To understand the principles of steam engine and Steam & Gas Turbine: **K2**

List of Experiments: (At least 8 of the following)

1. Study of Fire Tube boiler.
2. Study of Water Tube boiler.
3. Study and working of Two stroke petrol Engine.
4. Study and working of Four stroke petrol Engine.
5. Determination of Indicated H.P. of I.C. Engine by Morse Test.
6. Prepare the heat balance sheet for Diesel Engine test rig.
7. Prepare the heat balance sheet for Petrol Engine test rig.
8. Study and working of two stroke Diesel Engine.
9. Study and working of four stroke Diesel Engine.
10. Study of Velocity compounded steam turbine.
11. Study of Pressure compounded steam turbine.
12. Study of Impulse & Reaction turbine.
13. Study of steam Engine model.
14. Study of Gas Turbine Model.

BME452- MANUFACTURING PROCESSES LAB

CO-1 : Students will be able to make the component using casting and finishing methods. **K2**

CO-2 : Students will be able to make the component using metal cutting / unconventional machining methods. **K2**

CO-3 : Students will be able to make the component using metal joining processes. **K2**

List of Experiments: (At least 8 of the following along-with study of the machines/processes)

1. Making of Pattern (Wax / Wooden)
2. Preparation of Mould and Casting
3. Shear-angle determination (using formula) with tube cutting (for orthogonal) on lathe machine.
4. Bolt (thread) making on Lathe machine.
5. Tool grinding (to provide tool angles) on tool-grinder machine.
6. Gear cutting on Milling machine.
7. Finishing of a surface on surface-grinding machine.
8. Drilling holes on drilling machine and study of twist-drill.
9. Study of different types of tools and its angles & materials.
10. Experiment on tool wear and tool life.
11. Experiment on jigs/Fixtures and its uses.
12. Gas welding experiment.
13. Arc welding experiment.
14. Resistance welding experiment.
15. Soldering & Brazing experiment.
16. Experiment on Unconventional Machining (any one among - Laser Cutting, CO2 Cutting, ECM, EDM etc.)

BME453- COMPUTER AIDED MACHINE DRAWING-II LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand and apply 3D software to develop a part model	**K3**
CO 2	Understand conventional representation of machine components, and welded joints	**K2**
CO 3	Understand and apply the basis of fit or limit system	**K3**
CO 4	Understand about Plummer Block Bearing, Machine Vice, Screw Jack, Engine Stuffing box.	**K2**
CO 5	Create 3D part models and assemblies of various machine components	**K3**
	DETAILED SYLLABUS	**0-0-2**

Unit I

Introduction to Part Modelling: Introduction to part modelling of simple machine components using any 3D software (like CATIA, PRO E, UGNX, Autodesk Inventor or SOLIDWORKS) covering all commands/ features to develop a part model (Minimum 24 machine components need to be developed).

Unit II

Introduction: Conventional representation of machine components and materials, Conventional representation of surface finish, Roughness number symbol, Symbols of Machine elements and welded joints. Classification of Drawings: Machine drawings, Production drawing, part drawing and assembly drawing. Introduction to detail drawing and bill of materials (BOM).

Unit III

Limits, Fits and Tolerances: General aspects, Nominal size and basic dimensions, Definitions, Basis of fit or limit system, Systems of specifying tolerances, Designation of holes, Shafts and fits, commonly used holes and shafts. List of Standard Abbreviation used.

Unit IV

Part Modelling & Assemblies of: Plummer Block Bearing, Machine Vice, Screw Jack, Engine Stuffing box, Lathe Tailstock, Feed Check Valve and Rams Bottom Safety Valve.

Note: All drawing conforms to BIS Codes.

Books and References:

1. Textbook of Machine Drawing, K C John, PHI.
2. Machine Drawing by K.R. Gopalakrishna, Subhas Stores.
3. A Textbook of Machine Drawing by PS Gill from S.K. Kataria& Sons.
4. Machine Drawing-KL Narayana, P Kannaiah, KV Reddy, New Age publications.
5. Engineering Graphics with AutoCAD, Bethune, PHI.
6. Machine Drawing, N. Siddeshswar, P Kannaiah, VVS Shastry, Tata McGraw Hill.
7. Fundamentals of Machine Drawing, Dr Sadhu Singh & P L Shah, Prantice Hall India.
8. Autodesk Inventor by Examples, Sam Tikoo, Wiley.